WAVE OPTICS

JEE Main & Advanced Revision Formula Sheet

PHYSICS FORMULA BOOKLET

1. ELECTROMAGNETIC NATURE OF LIGHT

- **Sinusoidal Wave Equations:**

$$E_y = E_0 \sin(kx \mp \omega t) \quad | \quad B_z = B_0 \sin(kx \mp \omega t)$$

Light is a transverse non-mechanical EM wave. E & B are in phase.

- **Speed of Light in Medium:**

$$v = \frac{1}{\sqrt{(\mu \varepsilon)}} = \frac{c}{\mu} \Rightarrow E_0 = v B_0$$

- **Refractive Index (μ):** $\mu = \frac{c}{v} = \frac{\lambda_{vac}}{\lambda_{med}} = \sqrt{(\mu_r \varepsilon_r)} \approx \sqrt{(\varepsilon_r)}$ [Added]
- **Intensity of EM Wave:** [Added]

$$I = \frac{1}{2} \varepsilon v E_0^2 = \frac{1}{2} c \varepsilon_0 E_0^2 \text{ (in vacuum)} \Rightarrow I \propto A^2$$

2. WAVEFRONTS & HUYGENS' PRINCIPLE

- **Wavefront:** Locus of all medium particles oscillating in the same phase. Light rays are always *perpendicular* to wavefronts.
- **Shapes of Wavefronts:**

Source Type	Wavefront Shape	Intensity Relation
Point Source	Spherical	$I \propto 1/r^2$ $(A \propto 1/r)$
Line Source	Cylindrical [Added]	$I \propto 1/r$ $(A \propto 1/\sqrt{r})$
Distant Source	Plane	$I = \text{const}$ $(A = \text{const})$

- **Secondary Wavelets:** Every point on a wavefront acts as a secondary spherical wavelet source traveling at speed v.

3. PRINCIPLE OF INTERFERENCE

Superposition of two coherent waves: $y_1 = A_1 \sin(\omega t - kx)$, $y_2 = A_2 \sin(\omega t - kx - \phi)$

Resultant Amplitude & Phase:

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos\phi$$

$$\tan\alpha = \frac{A_2 \sin\phi}{A_1 + A_2 \cos\phi}$$

Resultant Intensity Equation:

$$I = I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos\phi$$

- **Phase Diff (ϕ) vs Path Diff (Δx):**

$$\phi = \frac{2\pi}{\lambda} \Delta x \quad | \quad \Delta x = \frac{\lambda}{2\pi} \phi$$

- **Coherent vs Incoherent Sources:**
  - *Coherent:* Constant ϕ with time. $I = I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos\phi$
  - *Incoherent:* Random $\phi(t)$. $I_{net} = I_1 + I_2$ (No stable fringes)
- **Slit Width & Intensity Relation:**

$$\frac{W_1}{W_2} = \frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$$

4. MAXIMA & MINIMA CONDITIONS

Constructive Interference (Maxima):

- $\phi = \pm 2n\pi$ (n = 0, 1, 2, ...)
- Path Diff: $\Delta x = \pm n\lambda$
- Max Amplitude: $A_{max} = A_1 + A_2$
- Max Intensity: $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$

Destructive Interference (Minima):

- $\phi = \pm (2n - 1)\pi$ (n = 1, 2, 3, ...)
- Path Diff: $\Delta x = \pm (2n - 1)\frac{\lambda}{2}$
- Min Amplitude: $A_{min} = |A_1 - A_2|$
- Min Intensity: $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Special Case (Identical Sources $I_1 = I_2 = I_0$):

$$I = 4I_0 \cos^2\frac{\phi}{2} \quad | \quad I_{max} = 4I_0, \; I_{min} = 0$$

Contrast Ratio & Visibility:

$$\frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(\sqrt{r} + 1)^2}{(\sqrt{r} - 1)^2} \text{ where } r = \frac{I_1}{I_2}$$

Fringe Visibility: $V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$ [Added]

5. YOUNG'S DOUBLE SLIT SETUP (YDSE)

Setup: Slit separation d, Screen distance D ($D \gg d$).

- **Path Difference at Angle θ:**

$$\Delta x = d \sin\theta \approx d \tan\theta = \frac{yd}{D} \text{ (for small } \theta)$$

- **Position of nth Bright Fringe (Central Max at $y = 0$):**

$$y_n = \frac{n \lambda D}{d} \quad (n = 0, \pm1, \pm2, ...)$$

- **Position of mth Dark Fringe:**

$$y_m = (2m - 1)\frac{\lambda D}{2d} \quad (m = 1, 2, 3, ...)$$

- **Linear Fringe Width (β):**

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$$

  - Fringe width is constant for all orders in YDSE.
  - Bright & dark fringes are equally spaced.
- **Angular Fringe Width (θ_0):**

$$\theta_0 = \frac{\beta}{D} = \frac{\lambda}{d} \text{ (in radians)}$$

- **YDSE in Medium of Refractive Index μ:** [Added]

$$\lambda' = \frac{\lambda}{\mu} \Rightarrow \beta' = \frac{\beta}{\mu}$$

WAVE OPTICS (CONTD.)

Optical Path, Thin Films, Diffraction & Polarization

6. OPTICAL PATH & GLASS SLAB SHIFT

- **Optical Path Length:** Distance in vacuum equivalent to path t in medium: $L_{opt} = \mu t$.
- **Path Difference Introduced by Thin Sheet:**

$$\Delta x_{extra} = (\mu - 1)t$$

- **Shift of Fringe Pattern (y_0):**

$$y_0 = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$$

  - Shift occurs towards the side where the slab is placed.
  - Fringe width β remains UNCHANGED.
- **Number of Fringes Shifted (N):** [Added]

$$N = \frac{y_0}{\beta} = \frac{(\mu - 1)t}{\lambda}$$

7. OBLIQUE YDSE & THIN FILMS [ADDED]

- **Oblique Incidence in YDSE:**
  If light strikes slits at angle α:

$$\Delta x = d(\sin\theta \mp \sin\alpha)$$

  Central maximum shifts to $\theta = -\alpha$.
- **Thin Film Interference (Thickness t, Refractive Index μ):**
  Reflected Light (includes π phase flip at denser boundary):
  - Maxima (Bright): $2\mu t\cos r = (2n - 1)\frac{\lambda}{2}$
  - Minima (Dark): $2\mu t\cos r = n\lambda$

  Transmitted Light (No extra phase flip):
  - Maxima (Bright): $2\mu t\cos r = n\lambda$
  - Minima (Dark): $2\mu t\cos r = (2n - 1)\frac{\lambda}{2}$
- **Wedge-Shaped Thin Film Fringe Spacing:**

$$\beta = \frac{\lambda}{2\mu\theta}$$ (θ in radians)

8. SINGLE SLIT DIFFRACTION

Slit width = a (or d), Screen distance = D.

- **Intensity Formula:**

$$I(\theta) = I_0\frac{\sin^2\alpha}{\alpha^2} \text{ where } \alpha = \frac{\pi a\sin\theta}{\lambda}$$

- **Condition for Minima (Dark Points):**

$$a\sin\theta = \pm n\lambda$$ (n = 1, 2, 3, ... ; n ≠ 0)

- **Condition for Secondary Maxima:**

$$a\sin\theta \approx \pm(n + \frac{1}{2})\lambda$$ (n = 1, 2, 3, ...)

- **Width of Central Maximum:**
  - Angular Width: $W_\theta = 2\theta_1 = \frac{2\lambda}{a}$
  - Linear Width on Screen: $W = \frac{2\lambda D}{a}$
- **Angular Width of Secondary Maxima:**

$$\theta_s = \frac{\lambda}{a}$$ (half the central max width)

- **Intensity Ratio of Peak Maxima:** [Added]

$$I_0 : I_1 : I_2 : I_3 = 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} \ldots \approx 1 : 0.045 : 0.016$$

9. CIRCULAR APERTURE & RESOLUTION

- **Airy Disc (Circular Aperture of Diameter d):**
  Angular radius of first dark ring:

$$\theta_R = 1.22\frac{\lambda}{d}$$

  Linear radius on focal plane of lens (f):

$$r = 1.22\frac{\lambda f}{d}$$

- **Rayleigh's Criterion for Resolution:**
  Two point objects are just resolved when central maximum of one falls on first minimum of the other.
- **Telescope Resolution Limits:**
  - Limit of Resolution: $d\theta = 1.22\frac{\lambda}{d}$
  - Resolving Power: $RP = \frac{1}{d\theta} = \frac{d}{1.22\lambda}$
- **Microscope Resolving Power:** [Added]
  - Limit of Resolution: $\Delta x = \frac{\lambda}{2\mu\sin\theta}$
  - Resolving Power: $RP = \frac{2\mu\sin\theta}{\lambda} = \frac{2NA}{\lambda}$

$NA = \mu\sin\theta$ is Numerical Aperture.

10. DIFFRACTION GRATING [ADDED]

- **Grating Element ($d = a + b$):**

$$d\sin\theta = n\lambda$$ (Principal Maxima)

- **Dispersive Power:** $\frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$
- **Resolving Power:** $RP = \frac{\lambda}{d\lambda} = nN$ (N = total lines)

11. POLARIZATION & PHOTOMETRY

- **Unpolarized Light through Polarizer:**

$$I_{out} = \frac{I_0}{2}$$ (Intensity halved always)

- **Malus' Law (Plane Polarized Light through Analyzer):**

$$I = I_0\cos^2\theta$$

θ = angle between transmission axes of polarizer & analyzer.

- **Two Polaroids in Series:** $I_2 = I_1\cos^2\theta'$
- **Brewster's Law (Polarization by Reflection):**

$$\tan\theta_p = \mu \Rightarrow \theta_p = \tan^{-1}(\mu)$$

  - At Brewster's angle $i = \theta_p$, reflected ray is 100% plane polarized.
  - Reflected & refracted rays are perpendicular: $\theta_p + r = 90°$.
- **Polarization Types** [Added]:
  - *Circular:* Superposition of 2 perpendicular equal-amplitude waves with $\Delta\phi = \pi/2$.
  - *Elliptical:* Superposition with unequal amplitudes or arbitrary $\Delta\phi$.
- **Optical Activity & Specific Rotation:**

$$[\alpha]_t^{\circ}C^{\lambda} = \frac{\theta}{L \times C}$$

θ = rotation angle (°), L = path length in decimeters (dm), C = concentration in g/cm³.